

RAJESH RAMANI

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PROFILE SUMMARY

Apple Alum and Agentic AI Engineering Consultant with a background in enterprise program leadership and hands-on AI product development. After years leading large-scale infrastructure programs, I deliberately transitioned into agentic AI, spending the past several months building and shipping production AI systems end to end. Currently an investor and Agentic AI Product Consultant for a hyperlocal creator marketplace startup, where I built **MatchScout**, an AI recommender that matches businesses with creators and is evaluated against real gig outcomes. Creator of the open-source **HPRC Framework**, an AI-native server-side templating engine. Experienced at bridging business, product, engineering, and AI teams, turning ambiguous problems into practical solutions with measurable impact.

ENGINEERING CENTERPIECES

HPRC Framework — Open-Source AI-Native Templating Engine

Python · Apache-2.0 · asyncio · Pydantic v2 · FastAPI · pytest GitHub: github.com/HPRCFramework/hprc-framework · Web: hprcframework.dev

- **Project Overview:** Designed and built HPRC end-to-end — a server-side templating engine that lets developers embed LLM prompts directly inside HTML and runs them automatically when the page is rendered.
- **Render Pipeline:** Figures out which prompts depend on which others, runs the independent ones concurrently, and caches the results so repeat renders are fast.
- **Multi-Provider LLM Layer:** Six interchangeable adapters (OpenAI, Anthropic, Gemini, Ollama/LM-Studio, Mock, and a multi-provider router) behind one simple interface. Each provider's SDK is loaded only when used.
- **Safe Extension Points:** External rules and allowlisted tool calls expose seams for customization without an expression language inside templates, so templates can't be turned into arbitrary code.
- **Architecture:** Clean layered design with Pydantic v2 throughout and a tolerant HTML parser.
- **Language Specification:** Authored the SPREP language specification (a mini-RFC) that defines the embedded-prompt syntax.

MatchScout — Onemyle Marketplace AI Recommender (PoC)

LangGraph · Claude Sonnet 4.6 · sentence-transformers · SQLite · Streamlit · LangSmith · pytest · GitHub Actions GitHub: [Onemyle-matchscout](https://github.com/Onemyle-matchscout)

- **Pipeline Architecture:** Built the entire recommender as a single LangGraph state machine: a rules-based filter narrows candidates, a sentence-transformer cosine ranker scores them, and a treatment-arm router decides whether the final pick is made by the rules baseline or by Claude using past-gig metrics and a versioned prompt.
 - **Outcome-Based Evaluation:** 50% of gigs flow through a no-LLM control, 50% through the Claude re-ranker, and per-arm quality is measured on real downstream gig outcomes with p-value significance testing. No synthetic golden set; no LLM-as-judge.
 - **Prompts as Tuning Surface:** Versioned prompt files, every recommendation tagged with the prompt version, dashboard segments quality and cost by version so prompt iteration is empirically driven.
 - **Observability:** LangSmith tracing on every LLM call. Trace URLs persisted alongside each recommendation so a single click in the dashboard opens the corresponding trace.
 - **Cost Tracking:** Per-call USD cost computed from token usage and persisted on each recommendation, enabling cost-per-arm and cost-per-prompt-version comparison.
 - **CI:** Smoke test on every push via GitHub Actions, exercising the no-LLM path (no API key required).
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AI / ML SKILLS

- **LLM APIs:** Anthropic Claude (sonnet-4.6, opus-4.7) · OpenAI GPT-4o · Google Gemini · function calling · tool_use · streaming · prompt caching · extended thinking
 - **Agent Frameworks:** LangGraph (StateGraph, Send API, HITL, MemorySaver, SqliteSaver) · LangChain (LCEL Runnables, ChatPromptTemplate, RunnableParallel/Branch/Passthrough/Lambda) · CrewAI (parallel + hierarchical) · Claude Agent SDK · Mem0
 - **Protocols:** Model Context Protocol (MCP) — tools, resources, prompt templates, streamable-http; FastMCP server + agentic clients
 - **Agentic Patterns:** Tool-use loops · ReAct · reflection / self-critique (Reflexion) · orchestrator-worker · hub-and-spoke · multi-agent crews · subagent decomposition · arm-routed treatment design
 - **RAG:** Embeddings (sentence-transformers, OpenAI text-embedding-3) · RecursiveCharacterTextSplitter separator hierarchy · cosine similarity · L2-normalization → dot-product optimization · citation grounding · metadata propagation through chunkers
 - **Vector Stores:** ChromaDB (Ephemeral + Persistent) · Qdrant
 - **Evaluation:** Outcome-based two-arm A/B with p-value significance · per-call USD cost tracking from token usage · LLM-as-judge bias awareness (position / verbosity / self-preference / style / drift / preference leakage) · gold-set construction · structured-output grading
 - **Observability:** LangSmith (@traceable decorator, span hierarchy, trace-URL drill-down from dashboards) · LangFuse (in progress)
 - **Validation:** Pydantic v2 · TypedDict · structured output
 - **Testing & CI:** pytest · GitHub Actions
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PROFESSIONAL EXPERIENCE

Investor / Agentic AI Product Consultant — Onemyle (Hyperlocal Creator/Business Marketplace Startup)

Remote · Mar 2026 – Present

- **AI Strategy & Architecture:** Lead AI strategy, technical architecture, and product direction for the startup. Currently designing and shipping **MatchScout** (see Engineering Centerpieces).
- **Onemyle User Chatbot:** Designed an AI-powered local-area assistant for Indian cities — a 3-node deployment with a local Gemma orchestrator, ChromaDB semantic cache, multi-LLM parallel calls (Claude · GPT-4o · Gemini), intent-based routing, and a FastAPI + React stack.
- **Product Direction:** Strategic input on roadmap, ICP refinement, AI product positioning, and category taxonomy design.

Principal Program Manager — Oracle Cloud Infrastructure (OCI)

Fremont, CA · Jan 2026 – Mar 2026

- **Fiber Supply Remediation Program:** Established and led a cross-functional program to resolve bulk fiber supply constraints impacting hyperscale data center builds.
- **Supplier Diversification:** Expanded fiber supplier base from 3 to 7 vendors, reducing supply risk and improving resilience for large-scale data center deployments.
- **Planning Tool:** Developed a lightweight bulk-fiber estimation tool and defined processes enabling colocation providers to leverage Oracle supply allocations.

Manager, Technical Program Management — Apple Inc.

Infrastructure Programs · Software Tools for PMO and DC Operations · Sunnyvale, CA · Apr 2011 – Jan 2024

- **Program Leadership:** Led multi-year infrastructure programs across compute, storage, network, on-prem cloud, big-data, and database environments at hyperscale, ensuring quality, consistency, and integrity across cost-optimization, automation, and cross-functional collaboration initiatives.
- **Data-Driven PMO Tooling:** Built and shipped a custom PMO portal and a DC operations prioritization tool (DCCP) that drove a **50% efficiency gain** in cross-org program tracking and automated reporting.

- **Budgeting & Capital Planning:** Managed capital budgets exceeding **\$20M** for large-scale infrastructure programs; established CapEx tracking mechanisms and partnered with finance and vendor management on right-sizing and optimization.
 - **Cost Savings & Consolidation:** Achieved multi-million-dollar cost savings by consolidating 400+ applications from high-cost data centers to Apple-owned facilities; drove decommissioning programs to recover power, space, and cost.
 - **Security & Compliance:** Met PCI security standards for critical retail applications by establishing **PCI-compliant High-Secure Environments (HSEs)** in collaboration with compliance, information security, and network/systems architecture teams.
 - **Cross-Functional Partnerships:** Network Engineering · Platform/Cloud SRE · Database Engineering · Storage Engineering · DC Operations · Apple Online Store · Apple Pay · Retail Apps · iCloud · Applied Machine Learning · IT · Security · Finance.
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PRIOR EXPERIENCE SUMMARY

- **Senior Project Manager, Site Operations and Information Security, PayPal (Aug 2010 – Feb 2011) (Contract)** Implemented information security solutions in collaboration with PayPal's Information Risk and Security organization (IRM), data center site operations, and platform SRE teams. Protected digital assets through **DDoS prevention, hardware security modules (HSMs), bastion infrastructure, and real-time traffic analysis systems for fraud detection and prevention.**
 - **Delivery Manager, IT Infrastructure Services, EDS/HP Enterprise Services, Chennai, India (2007–2010)** Led a 125+ member team providing 24/7 L1/L2 incident management and service delivery across infrastructure services, including compute, backups, identity and access management, and storage.
 - **Release Manager, Bank of America, Agoura Hills, CA (2006–2007)** Managed and coordinated the deployment of releases for a loan origination system development project.
 - **Chief Technology Officer, iDrive Inc., Woodland Hills, CA (1998–2006)** Designed and led the development of a commercially successful data backup product, driving technical innovation and product success.
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CAPSTONE PROJECTS

Built across a focused 4-month agentic AI curriculum to demonstrate framework breadth and pattern coverage.

- **StockSage MCP Server** (*FastMCP · Claude · ChromaDB · sentence-transformers*) — MCP server with 4 stock-analysis tools over streamable-http. Generic agentic loop dispatches tools dynamically with zero hardcoding. Persistent ChromaDB vector store for earnings-transcript embeddings; MCP prompt templates decoupled from client.
 - **ReviewCrew** (*CrewAI · GitHub REST API · Claude*) — 2-agent crew (code reviewer + security reviewer) analyzing GitHub PRs in parallel. Custom GitHub tool integration; human-in-the-loop approval before comment posting.
 - **ResearchBot** (*LangGraph · Kafka · DuckDuckGo*) — Multi-step web-research agent on a LangGraph state graph with typed state, conditional edges, and checkpointing. Human-in-the-loop interrupt and resume for mid-graph approval.
 - **DocTalk** (*LangChain · Qdrant · sentence-transformers · Claude*) — End-to-end RAG pipeline over PDFs: ingestion, chunking, embedding, cosine retrieval, and cited answer generation. Per-page metadata propagation through the chunker enables citation grounding.
 - **ToolBot** (*asyncio · Pydantic v2 · Anthropic / OpenAI / Gemini*) — Unified multi-LLM CLI agent behind a single tool-use loop. Streaming, multi-turn memory with sliding-window truncation, provider-specific dialect handling, Pydantic v2 schema validation.
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EDUCATION & CERTIFICATIONS

- M.S., Information Systems — California State University, Los Angeles
- B.S., Computer Science — National Institute of Technology, India
- **AWS Solutions Architect — Associate · KCNA · PMI-PMP · Certified ScrumMaster · GCP Cloud Digital Leader**

LINKS

- LinkedIn: linkedin.com/in/rvrajesh123
- GitHub org (AI projects): github.com/Agentic-and-AI-Engineering-Projects
- HPRC Framework: github.com/HPRCFramework/hprc-framework · hprcframework.dev